BP-R Stage 3: approved priority and static contract

6 October 2026 — source/static review; S3-G3 disposition pending

Status. Reference route A, upstream admission M.A and fixed junction priority N.A are approved. The unavailable third source is excluded from exact replication. The equations below specify interface accounting and local bounds; they do not certify a completed reference solver, a transient compact closure, clearance, control benefit or a gradient.

1. The approved receiving allocation

All quantities here are rates in vehicle equivalents per second. $D_r \geq 0$ is ramp sending demand at the merge; $D_m \geq 0$ is the sum of compatible non-ramp incoming demands to the same shoulder-lane receiver; $R \geq 0$ is the available receiving rate consistent with the traffic field and existing moving constraints. $p \in (0,1)$ is a fixed physical ramp priority, common across models and control schedules. It is not the upstream metering rate.

The selected conservative node operator is

$$(f_r, f_m) = \begin{cases} (D_r, D_m), & D_r + D_m \leq R, \\ \big(\min\{D_r, \max\{pR, R - D_m\}\},\ R - f_r\big), & D_r + D_m > R. \end{cases} \tag{1}$$

In the second branch the displayed first coordinate defines f_r before the second coordinate is evaluated. This follows the fixed-priority merge family in Daganzo's 1994 network CTM original, Section 3.2. Grouping non-ramp inflows and joining moving constraints are declared task adaptations.

Under scarcity, $D_r > R - D_m$ and $pR \in [0, R]$. Hence

$$\max\{0, R - D_m\} \leq f_r \leq \min\{D_r, R\}, \qquad 0 \leq f_m = R - f_r \leq D_m. \tag{2}$$

Together with the unconstrained branch,

$$f_r + f_m = \min\{D_r + D_m, R\}, \qquad 0 \leq f_b \leq D_b \quad (b = r, m). \tag{3}$$

Thus receiving is spent once and unused shares are reallocated. If both streams can use their contested shares, the allocation is pR and $(1-p)R$.

If compatible non-ramp movements have already budgeted requests d_j, $D_m = \sum_j d_j$, their subdivision is

$$f_j = \frac{d_j}{D_m} f_m \quad (D_m > 0); \qquad f_j = 0 \quad (D_m = 0). \tag{4}$$

This preserves request bounds and sums to f_m. Donor budgets must be formed before subdivision; independent full-capacity requests are not admissible. Other receiving lanes keep their source local allocation.

2. Upstream admission and population ownership

W^k is waiting mass upstream of the finite ramp approach, A^k is offered arrival mass in a time bin, r^k is the common permission rate, R_a^k is approach receiving and Δt is the bin duration. A reviewable discrete boundary transcription is

$$B^k = \min\{W^k + A^k,\ r^k \Delta t,\ R_a^k \Delta t\}, \qquad W^{k+1} = W^k + A^k - B^k. \tag{5}$$

Admission B^k is credited once to the ramp. It is not actual merge entry. Stored approach traffic can enter the freeway faster than the contemporaneous upstream command. All models must retain that propagation and count W.

For road/ramp control volumes c, let n_c^k be continuous mass. Each internal transfer F_{cb}^k is debited from c and credited to b. Let I^k include all offered external arrival mass, and E^k include all sink exit mass. For total waiting plus road/ramp inventory M^k,

$$M^k = W_r^k + W_m^k + \sum_c n_c^k, \qquad M^{k+1} - M^k = I^k - E^k. \tag{6}$$

Here W_m includes any blocked mainline entrance population. Every internal debit/credit cancels. Conservation alone does not establish receiving/storage correctness; those also require the coupled flux law.

The supplied journal's lateral recipe sends mass from (i, ℓ) to $(i+1, b)$. Counting a separate same-cell lateral transfer followed by another longitudinal transfer for that mass would change the source update. Descriptor insertion does not add mass to M.

For a particular actual entry/transfer stream, let $H(t)$ be its accumulated continuous mass plus a declared initial event progress. The deterministic source descriptor count can be expressed as

$$K(t) = \lfloor H(t) \rfloor - \lfloor H(0) \rfloor. \tag{7}$$

This selects unit-threshold descriptors; it does not round H, the flow or n_c. Event progress is public initial data, not an optimized phase. Threshold crossing times and coincident events remain part of the reference transcription.

The 2004 antecedent, Section 3, gives the sending-lane speed as birth speed. The final 2006 journal, Appendix A, then constrains realized motion using

$$v(t) = \min\{v_{\text{des}}(t), v_{\text{dwn}}(t)\}, \qquad a(v, x) = a_0(1 - v/v_{\max}) - gG(x). \tag{8}$$

v_{dwn} is the immediately downstream KW-stream speed, not the lane-choice look-ahead speed. Positions move with realized speed; desired speed advances with the source acceleration law. $a_0, v_{\max}, G$ are physical parameters, not functions assigned arbitrarily from permission r.

3. Complete system time and the remaining boundary

Let $A(t)$ and $E(t)$ be cumulative offered arrivals and exits from the declared start. For initial total population M_0,

$$M(t) = M_0 + A(t) - E(t), \qquad J_{\mathrm{TST}} = \int_0^{T_{\mathrm{clear}}} M(t)\,dt. \tag{9}$$

J_{TST} has units vehicle-equivalent seconds. Mean remaining system time is J_{TST}/P for common finite population P; pre-start elapsed time is a common constant. An asymptotic convergent inventory integral with a certified tail bound may replace finite T_{clear}. No extra merge-work cost is added.

For schedules a, b with common population, arrivals and initial inventory, extend each cleared exit history constantly to a common terminal time T:

$$J^a_{\mathrm{TST}} - J^b_{\mathrm{TST}} = -\int_0^T \big(E^a(t) - E^b(t)\big)\,dt. \tag{10}$$

Identical full exits therefore mean identical TST even if queues are relocated. Different mean capacity or one terminal count is not a complete-objective comparison. This identity is bookkeeping, not a feasible-control witness.

Clearance is not proved. Finite offered arrivals and a positive draining permission do not alone establish clearance of the adapted reference. The joined law must justify eventual discharge: reachable exits, available receiving and constraints that release or depart. A certified residual tail is an alternative; dropping remaining mass or assuming an empty road is not.

4. What the source evidence does and does not finish

The supplied journal and 2004 antecedent define traffic/event ingredients. The 2003 moving-boundary report gives a cumulative-count coupling route; its node/event integration here is not yet verified. A guessed acceleration-capacity penalty cannot replace that route.

Existing compact alternatives are serious. The 2011 endogenous merge work and 2016 moment manuscript predict mean capacity under their assumptions, not a matched transient state update for this upstream meter. The 2021 hysteretic CTM uses a different receiving/control construction. Importing any of these without qualification would confound representation with a changed task law.

An independent rival must initialize from its declared public projection and evolve its own speed/mode/moment state. It cannot borrow the reference's future entries or constraint history. Unknown accuracy of a bound law can be tested later; a missing law cannot be silently delegated to an experiment.

Result: partial/evidence-limited formulation. Local bounds and accounting are explicit. A complete joined reference/clearance and a strong independent transient compact closure remain unqualified. This is neither a necessity proof for finer state nor a sufficiency/impossibility theorem for compact models. Stage 3 disposition is pending S3-G3; Stage 4 remains unopened.

Source versions and links are recorded in `post_priority_evidence_audit.md`; the plain-language interpretation and remaining interfaces are in `joined_contract_and_reduction_audit.md`.